IMO 2026 — Problem 1

Solution by GPT-5.6 Sol (xhigh reasoning)

2026-07-18

Source: `results/gpt-5.6-sol-xhigh/problem-01/current.md` — autonomous run, no human intervention; solution as written by the model.

Problem

There are 2026 integers greater than 1 written on a blackboard, not necessarily different. In a move, Confucius chooses two integers $m > 1$ and $n > 1$ from different places on the blackboard and replaces these two integers with $\gcd(m, n)$ and $\frac{\operatorname{lcm}(m,n)}{\gcd(m,n)}$. He continues to make moves while it is possible to do so. (a) Prove that, regardless of the choices of Confucius, after finitely many moves, exactly one integer M on the blackboard is greater than 1. (b) Prove that the value of M does not depend on the choices of Confucius.

Solution document

Status

solved

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Approaches tried

- **Lexicographic descent (successful):** track the product of all board entries and the number of entries greater than 1. A move with nontrivial gcd decreases the product, while a move with gcd 1 preserves the product and decreases the number of nonunit entries. Details are in approaches/lexicographic-descent.md.
- **Prime-adic exponent invariant (successful):** for each prime p, the selected exponent pair (a, b) becomes $(\min(a, b), |a-b|)$; the gcd of all p-adic exponents is consequently invariant.

Details are in approaches/prime-adic-invariant.md, with the invariant isolated in lemmas/valuation-gcd-invariant.md.

- **Computational sanity check:** random plays on 5000 small initial boards (four plays per board) all terminated at the value predicted by the valuation formula; see code/random_verify.py. This check is not used in the proof.

Current best

The problem is solved. If the initial integers are $x_1, \dots, x_{2026}$, then every play terminates with exactly one nonunit, and its value is explicitly

$$M = \prod_p p^{\gcd(v_p(x_1),\dots,v_p(x_{2026}))},$$

where the gcd of an all-zero list is 0. Only finitely many factors differ from 1.

Full proof

Let the entries at any stage be $y_1, \dots, y_{2026}$, and define

$$P = \prod_{i=1}^{2026} y_i, \qquad K = \#\{i : y_i > 1\}.$$

All entries are positive integers throughout the process.

(a) Termination and the number of surviving nonunits

Consider a move involving $m, n > 1$, and put $d = \gcd(m, n)$. The product of the two new entries is

$$d \cdot \frac{\operatorname{lcm}(m,n)}{d} = \operatorname{lcm}(m,n) = \frac{mn}{d}.$$

All other entries are unchanged, so the new board product is P/d.

If $d > 1$, the positive integer P strictly decreases. If $d = 1$, the selected entries are replaced by

$$1 \quad \text{and} \quad mn.$$

In this case P is unchanged, while K decreases by exactly 1, because two entries greater than 1 have been replaced by precisely one entry greater than 1.

Suppose, for contradiction, that some sequence of moves were infinite. Along it, P would be a nonincreasing sequence of positive integers, so from some point onward P would be constant. Every move from that point onward would therefore have $d = 1$. But then the nonnegative integer K would decrease by 1 at every subsequent move, which cannot happen infinitely many times. Thus every sequence of moves is finite.

A move is possible exactly when $K \geq 2$. Hence, when Confucius can no longer move, $K \leq 1$. On the other hand, a move can never leave no entry greater than 1: if $d > 1$, its first output d is greater than 1, while if $d = 1$, its second output mn is greater than 1. Since initially $K = 2026 > 0$, we therefore have $K \geq 1$ at every stage. Consequently the terminal board has exactly one entry $M > 1$.

(b) Independence of the choices

Fix a prime p. For any positive integer z, let $v_p(z)$ denote the exponent of p in its prime factorization. We claim that

$$G_p = \gcd(v_p(y_1), \dots, v_p(y_{2026})) \qquad (1)$$

is invariant under every move. Here, as usual, $\gcd(0, r) = r$, and an all-zero gcd is defined to be 0.

Indeed, let

$$a = v_p(m), \qquad b = v_p(n)$$

for the two selected entries. Their two replacements have p-adic exponents

$$v_p(\gcd(m, n)) = \min(a, b)$$

and

$$v_p\left(\frac{\operatorname{lcm}(m, n)}{\gcd(m, n)}\right) = \max(a, b) - \min(a, b) = |a - b|.$$

If, without loss of generality, $a \le b$, then

$$\gcd(\min(a, b), |a - b|) = \gcd(a, b - a) = \gcd(a, b). \qquad (2)$$

The last equality holds because an integer divides both a and $b - a$ if and only if it divides both a and b. Thus replacing a, b by $\min(a, b), |a - b|$ does not change their gcd. Taking the gcd with all the unchanged exponents proves that (1) is invariant.

Now denote the initial entries by $x_1, \dots, x_{2026}$ and set

$$g_p = \gcd(v_p(x_1), \dots, v_p(x_{2026})).$$

By part (a), the terminal entries are M and 2025 copies of 1, in some order. Applying the invariant (1) at the initial and terminal boards gives

$$g_p = \gcd(v_p(M), 0, \dots, 0) = v_p(M) \qquad (3)$$

for every prime p. It follows from unique prime factorization that necessarily

$$\boxed{M = \prod_p p^{g_p} = \prod_p p^{\gcd(v_p(x_1), \dots, v_p(x_{2026}))}.}$$

If a prime divides none of the initial entries, its exponent in this product is 0, so the product has only finitely many nontrivial factors. The displayed value is determined solely by the initial board. Therefore the terminal value M is independent of all choices made by Confucius. $\square$